

Fig. 2: An actual-size, single-side PCB for microcontroller-based pulse counter

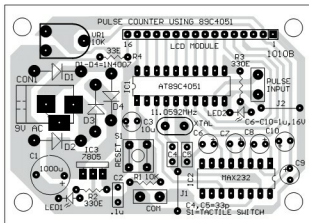


Fig. 3: Component layout for the PCB

P3.7 of the microcontroller. Port pins P1.0 through P1.7 of the microcontroller are connected to data pins D0 through D7 of the LCD, respectively. Port pins P3.4, P3.3 and P3.2 are connected to register-select RS, read-write R/W and enable E of the LCD, respectively.

The data is sent to the LCD in ASCII format for display. Only the commands are sent in hex form to the LCD. Register-select RS signal is used to distinguish between data (RS=1) and command (RS=0). Using preset VR1, you can control the contrast of the LCD. Resistor R4 limits the current through backlight of the LCD.

Port pin P3.1 of the microcontroller is used to interface with the PC's hyper terminal through RS-232 interface MAX232 IC (IC2). The microcontroller provides a transmit channel for serial data transfer. Transmit data

pin (TXD) is specified at port pin P3.1. The microcontroller is connected to T_{IN} (pin 11) of MAX232. T_{OUT} (pin 14) of IC2 is connected to pin 2 of the COM port connector. The signals provided on these pins are TTL-level and must be boosted and inverted through a MAX232 converter to comply with RS-232 standard.

The MAX232 has two internal charge pumps that convert +5V into $\pm 10V$ (unloaded) for RS-232 driver operation. The first converter uses capacitor C8 to double the +5V input to +10V on capacitor C10 at pin 2. The second converter uses capacitor C7 to invert +10V to -10V on capacitor C6 at pin 6.

All the serial communication is controlled through special-function register SCON. This register contains mode-selection bits, ninth data bit for transmit and receive (TB8 and RB8), and serial-port interrupt bits (TI and RI). Serial communication requires standard baud rate. Timer-1 is configured in auto-reload mode to generate baud rate. The baud rate is determined as:

$$\text{Baud rate} = \frac{(2^{\text{SMOD}} \times \text{Oscillator frequency})}{(32 \times 12 \times (256 - \text{TH1}))}$$

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC3). Capacitor C2 bypasses the ripples present in the regulated supply.



Fig. 4: Hyperterminal window for name and icon



Fig. 5: Hyperterminal window for COM port selection

LED1 acts as the power indicator and R2 limits the current through LED1.

Construction and testing

An actual-size, single-side PCB for the microcontroller-based pulse counter is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Connect the assembled circuit to the COM port of the computer. The accumulated pulse count transferred is sent to the PC through the COM port using the HyperTerminal program. To open the HyperTerminal program, go to Start → Programs → Accessories → Communications → HyperTerminal. You will see a window as shown in Fig. 4.

Type the desired name and click 'ok.' Select a COM port (refer Fig. 5) while ignoring the other options and click 'ok.'